



Exocentric Distance Experiment in Virtual Reality

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Introduction & Summary of the Original Paper

Key Question: Is the space we visually perceive intrinsically curved in a geometric sense?

- **Koenderink et al. (2000):** Used an *exocentric pointing* method to measure the curvature of visual space at the height of gaze.
 - Classical Geometric Basis: Used triangles and Gauss's angular excess method:
 - In curved space, triangle angle sums deviate from 180° .



Real-Life Setup

- Experiments conducted outdoors in natural settings (a field).
- Observers stood at the center of equilateral triangles and the Target sphere was placed on different distances (2m-20m) from the center of triangles
- Measured the angles by pointing towards “virtual” directions with minimal bodily movement but freedom to move head and eyes.

Real-Life Setup

Tripod Setup: The Observer was standing on the center of equilateral triangle observing the Pointer and the Target Sphere on opposite vertices. Observer allowed Eye, Head and body movement standing on fixed position. And can rotate the pointer using remote control.



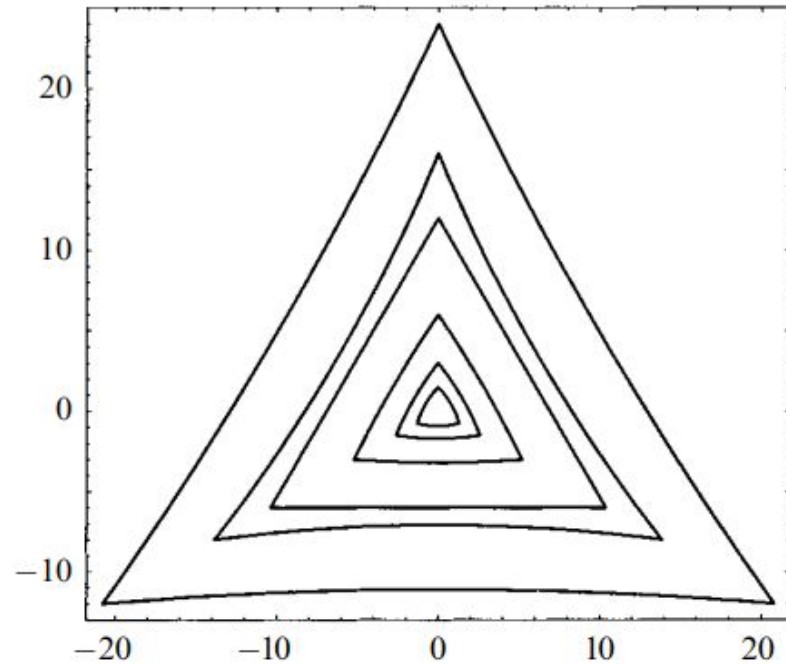


Description of the Original Results & Implications

Main Discovery: The curvature of visual space *varies* as a function of distance, not always negative (hyperbolic).

- Near distances: “Fat” triangles (sum of angles $>180^\circ$, elliptic curvature).
- Intermediate: Geometry close to Euclidean (sum $\approx 180^\circ$).
- Far distances: “Thin” triangles (sum $<180^\circ$, hyperbolic curvature).

Visual illustration of these “fat” and “thin” triangles along the depth axis from the observer.





Adaptation for VR

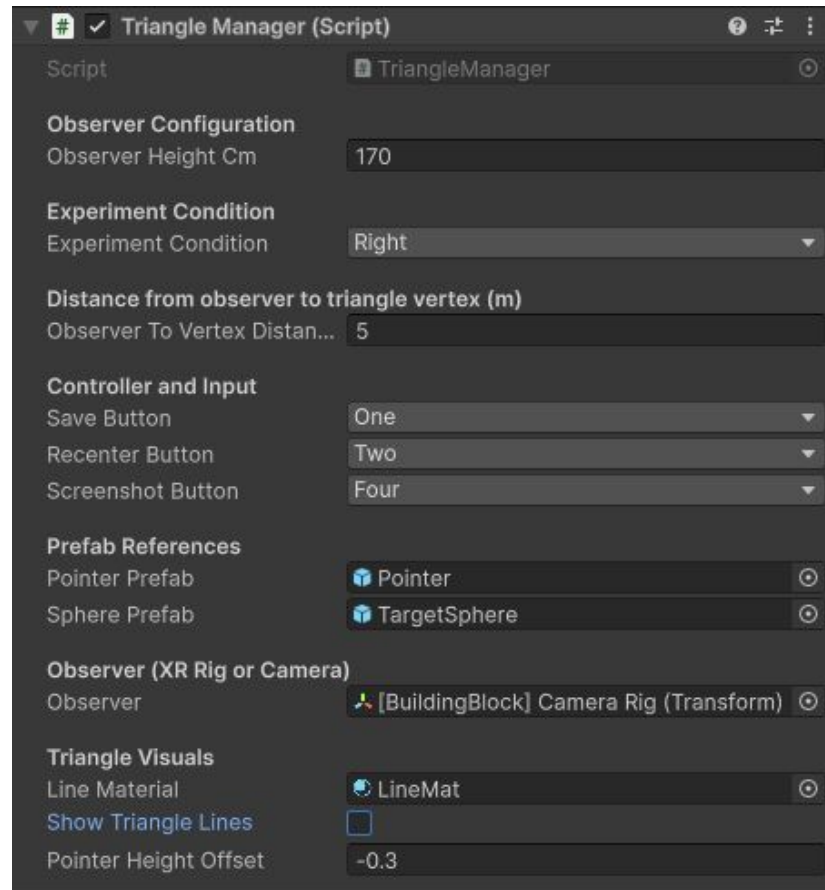
Virtual Environment: Implemented an Oculus-based VR replica of the physical triangle experiment.

Custom Unity Implementation:

- Builds equilateral triangles around a central observer in VR.
- Observer's eye height and triangle distance matched to real-life.
- "Pointer" and "sphere" objects mark triangle corners; participants rotate pointer to indicate perceived angle.
- Data logged and saved in CSV file for analysis.

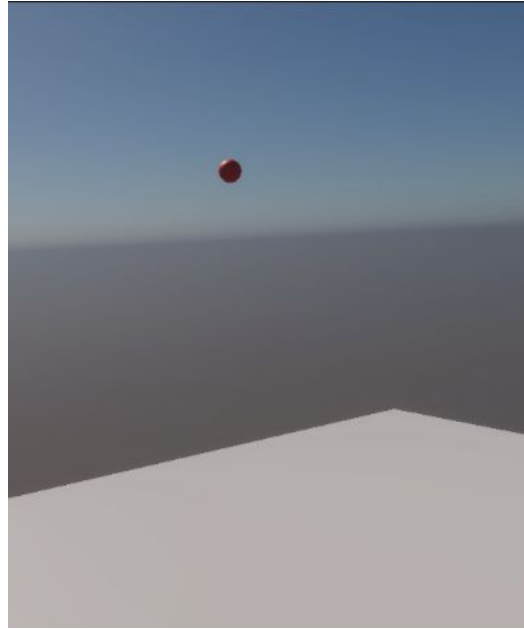
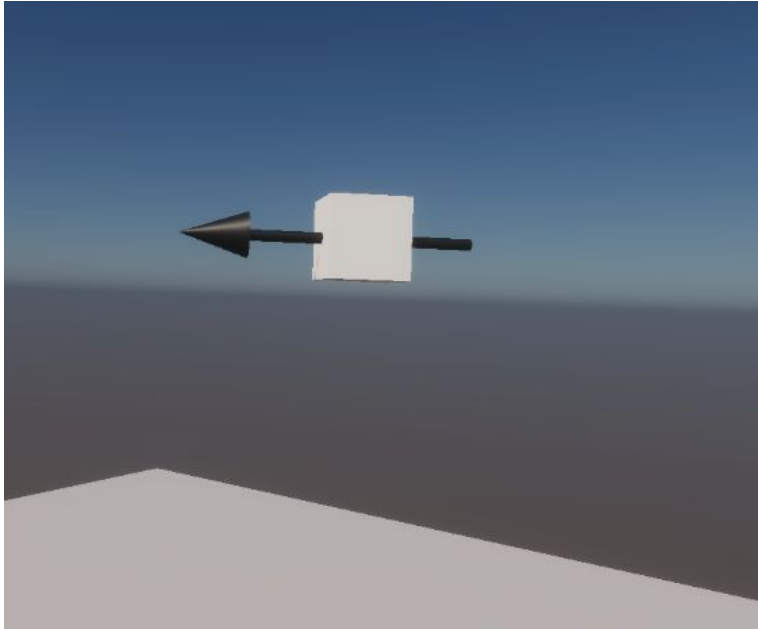
Virtual Environment in Unity

Managing Experimental Setup
through Triangle Manager
Script Inspector





Prefabs





Key Design Choices

- Rigid control of triangle geometry and distances to ensure consistency.
- Observer's real eye height measured and used in all trials for accurate perception alignment.
- Pointer and sphere objects are spawned exactly at opposite vertices of the triangle, aligned at eye height.
- At the start of each trial, the pointer is perfectly aligned to the target sphere, making a 0° angle with the triangle side (baseline).
- Participant is instructed to first point the pointer to themselves (establish reference), then to rotate the pointer towards the target sphere according to their perception.
- When the participant is satisfied with the pointer alignment, they press a button to save the current pointer angle relative to the initial triangle side.
- Recorded data stored in a CSV file including these attributes:
 - Distance (m)
 - Observer Height (cm)
 - Pointer Angle (degrees)
 - Condition (Left or Right triangle configuration)



Preliminary Results

- What was measured: For each distance (Observer to Target Sphere), how the subject's pointer deviated from the initial alignment (i.e., the perceived angle at triangle vertex).
 - Observed Trends:
 - Near distance (5m): Small positive pointer angles (matching “fat triangle”/elliptic).
 - Intermediate distance (10m): Geometry close to Euclidean (sum $\approx 180^\circ$).
 - Farther (15–20m): Larger negative pointer angles (matching “thin triangle”/hyperbolic).
- This pattern is consistent with Koenderink: positive excess near, negative deficit at distance.

Pilot Run Participant 1

Trail 1			
Distance_m	ObserverHeight_cm	PointerAngle_degrees	Condition
5	179.8	2.51	Right
5	179.8	2.22	Left
10	179.8	-0.77	Right
10	179.8	-0.61	Left
15	179.8	-7.47	Right
15	179.8	-9.88	Left
20	179.8	-13.03	Right
20	179.8	-14.5	Left

Trail 3			
Distance_m	ObserverHeight_cm	PointerAngle_degrees	Condition
5	179.8	2.47	Right
5	179.8	2.18	Left
10	179.8	-0.83	Right
10	179.8	-0.76	Left
15	179.8	-7.61	Right
15	179.8	-10.53	Left
20	179.8	-12.91	Right
20	179.8	-14.62	Left

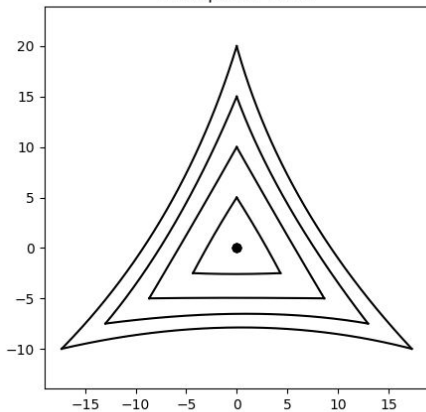
Trail 2			
Distance_m	ObserverHeight_cm	PointerAngle_degrees	Condition
5	179.8	2.95	Right
5	179.8	1.8	Left
10	179.8	-1.4	Right
10	179.8	0.1	Left
15	179.8	-8.67	Right
15	179.8	-9.57	Left
20	179.8	-11.5	Right
20	179.8	-16.13	Left

Trail 4			
Distance_m	ObserverHeight_cm	PointerAngle_degrees	Condition
5	179.8	2.21	Right
5	179.8	2.63	Left
10	179.8	-0.31	Right
10	179.8	-1.24	Left
15	179.8	-6.72	Right
15	179.8	-11.61	Left
20	179.8	-14.21	Right
20	179.8	-13.41	Left

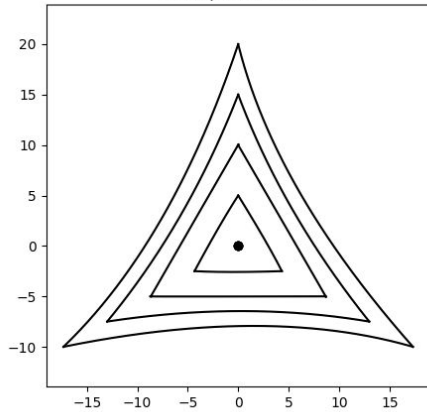


Pilot Run Participant 1

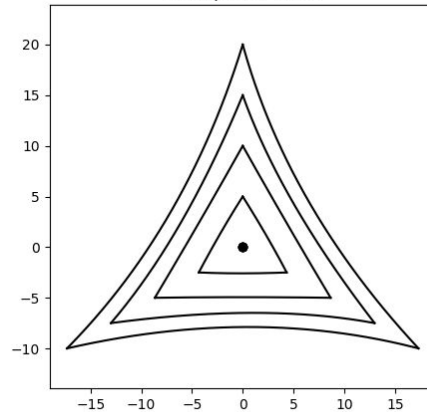
Participant 1 Trail 1



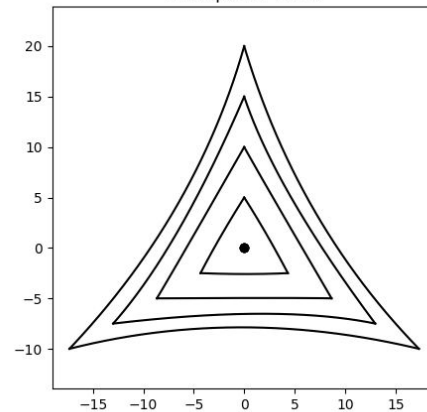
Participant 1 Trail 2



Participant 1 Trail 3



Participant 1 Trail 4



Pilot Run Participant 2

Trail 1			
Distance_m	ObserverHeight_cm	PointerAngle_degrees	Condition
5	173.7	2.41	Right
5	173.7	1.93	Left
10	173.7	-0.27	Right
10	173.7	-8.29	Left
15	173.7	-10.7	Right
15	173.7	-13.41	Left
20	173.7	-13.03	Right
20	173.7	-17.91	Left

Trail 3			
Distance_m	ObserverHeight_cm	PointerAngle_degrees	Condition
5	173.7	2.56	Right
5	173.7	1.99	Left
10	173.7	-1.04	Right
10	173.7	-0.05	Left
15	173.7	-7.68	Right
15	173.7	-10.61	Left
20	173.7	-13.16	Right
20	173.7	-15.88	Left

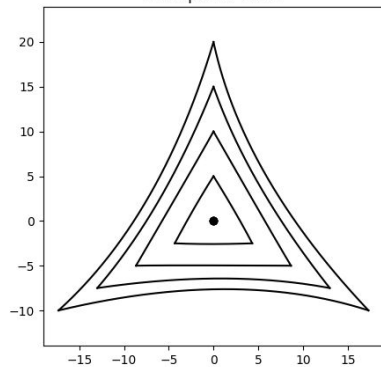
Trail 2			
Distance_m	ObserverHeight_cm	PointerAngle_degrees	Condition
5	173.7	1.63	Right
5	173.7	2.05	Left
10	173.7	-1.25	Right
10	173.7	-0.34	Left
15	173.7	-9.68	Right
15	173.7	-9.18	Left
20	173.7	-14.77	Right
20	173.7	-14.89	Left

Trail 4			
Distance_m	ObserverHeight_cm	PointerAngle_degrees	Condition
5	173.7	2.08	Right
5	173.7	1.87	Left
10	173.7	-0.94	Right
10	173.7	-1.06	Left
15	173.7	-8.77	Right
15	173.7	-10.72	Left
20	173.7	-13.26	Right
20	173.7	-16.88	Left

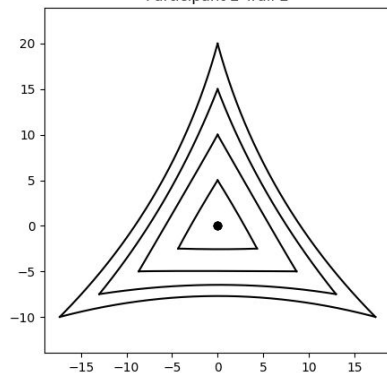


Pilot Run Participant 2

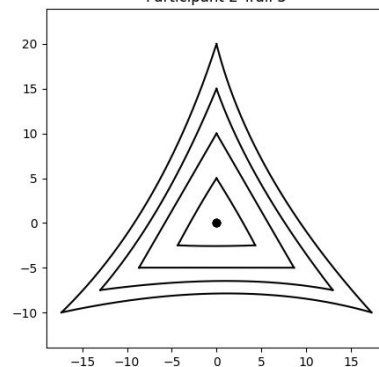
Participant 2 Trail 1



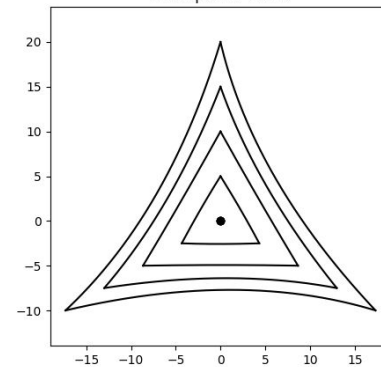
Participant 2 Trail 2



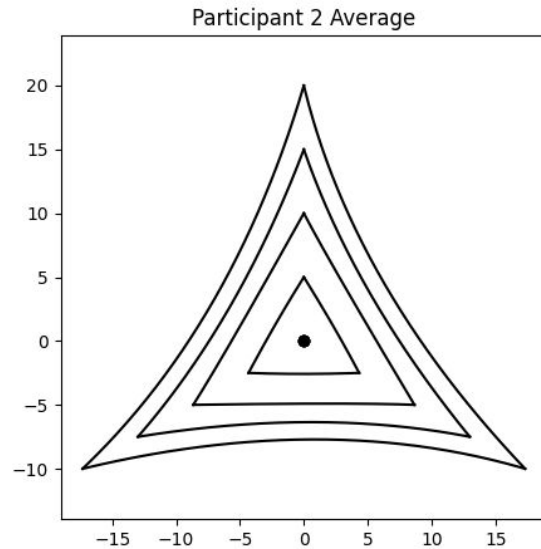
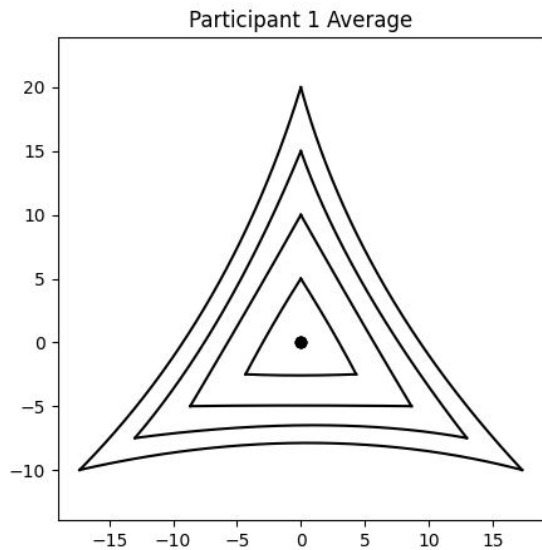
Participant 2 Trail 3



Participant 2 Trail 4



Inter-Observer Variability





Conclusion and Future Perspectives

Does VR Replicate Reality?

- Initial results show a clear *qualitative match* to the original: near/far space curvature effects are reproduced in immersive VR.
- But the Angle Deviation is smaller than the original as in the original experiment the angle was deviated upto 30 degrees.
- *Quantitative differences* might emerge with more participants and refined protocols.

What Next?

- Gather data from more subjects to characterize inter-individual variability.
- Test in a “naturalistic” VR scene (e.g., a field, as in the original, with rich depth cues).
- Explore *Augmented Reality (AR)* to test curvature in the user’s real environment with digital overlays for true ecological validity.



Reference

Koenderink, J. J., van Doorn, A. J., & Lappin, J. S. (2000).

Direct measurement of the curvature of visual space. *Perception*, 29(1), 69–79.

<https://doi.org/10.1068/p2921>